Bloch Theorem & Visualization of Energy Band

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Course: Numerical Analysis

I. Introduction

A. Motivation

這個報告主要是要以數值分析方法致敬電子雙狹縫實驗，事情要由學期初應用數學有提到一個叫做 Bloch Theorem 的物理定律，用來描述在一群排列整齊的晶格分子，提供週期性的位 能，此時電子們會表現出“一帶一帶”、即所謂的“能帶結構”。聽到這裡我想到了從前聽過得電子雙狹縫實驗，想到他們似乎都是起初有隨機分佈的概念，隨著樣本數變多之後會開始呈現出機率波的概念（如 Figure1[1]），因此我就發想這跟能帶似乎有異曲同工之妙(Figure2)，然後又讓我回想起 Monte Carlo method，都是要把樣本數拉高後就會開始出現很酷的結果，所以本篇報告就會期望以 Monte Carlo method 去做視覺化。

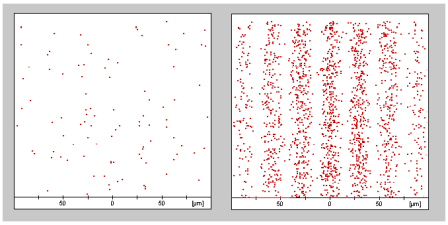

Figure 1 [1]

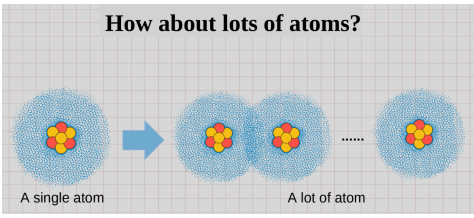

Figure 2

B. Bloch Theorem

But first we note that we consider the nearly free electron model, see down picture(left figure for free electron, V=0, right figure for nearly free electron, V $\approx$ 0)

Figure 3

Schrödinger's Equation with periodic potential energy:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x), V(x+a) = V(x)$$

Bragg Diffraction and inverse Fourier Transform give:

$$\psi_k(x) = \sum_G C_{k-G} e^{i(k-G)x}, \quad V(x) = \sum_G V_G e^{iGx}$$

And by above we obtain the following matrix, with main diagonal show the potential energy, remain are kinectic energy term.

$$\begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \iddots \\ \cdots & (k-G_1)^2 - E & V_1 & 0 & \cdots \\ \cdots & V_1 & (k-G_2)^2 - E & V_1 & \cdots \\ \cdots & 0 & V_1 & (k-G_3)^2 - E & \cdots \\ \iddots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} C_{k-G_1} \\ C_{k-G_2} \\ C_{k-G_3} \\ \vdots \end{pmatrix} = 0$$

Figure 4 [1]

II. Analysis

A. Arguements Setting

```python
def get_nfe_energy(k, a, V0, N_waves):
    """
    In NFE model, compute the eigenvalue in particular energy。
    """
```

k, a, V0, N_waves ⇒ 動量、晶格常數、位能、矩陣大小

B. Monte Carlo method

Note that[2]:

- Error $\propto \frac{1}{\sqrt{N}}$ as N is large enough (Central Limit Theorem).
- Truncation errors dominate (dependent on the choice of matrix size).

C. Some Visualization

The basic one(Left) bases on the soluitons of the matrix, no utilization of the Monte Carlo Method. Middle one is to show that the Monte Carlo method really works!

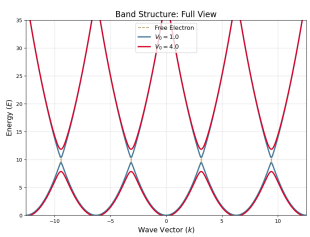

Figure 5 Matrix Solution

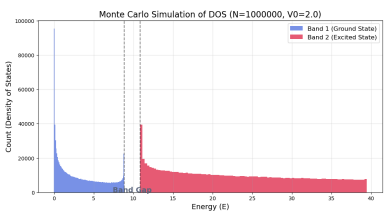

Figure 6

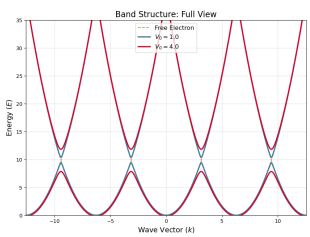

Figure 7 Matrix Solution

The following is to show the different N simple of Visualization(from non order to something, not clear to clear):

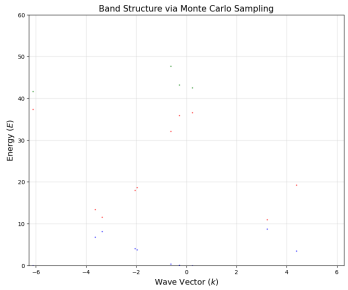

Figure 8

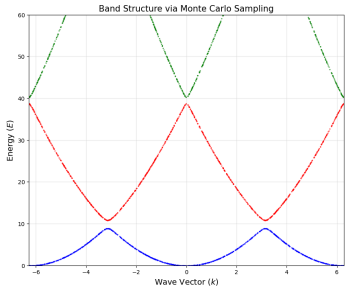

Figure 9

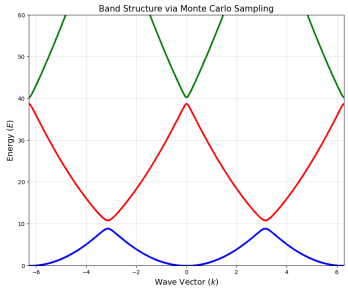

Figure 10

這裡可見樣本數多了起來那跟能帶的樣子也越來越像。

III. Results & Conclusion

A. Errors

Let's have some analysis about the convrgence.(The errors convrgence dramatically and really follow the theory, $\propto \frac{1}{\sqrt{N}}$)

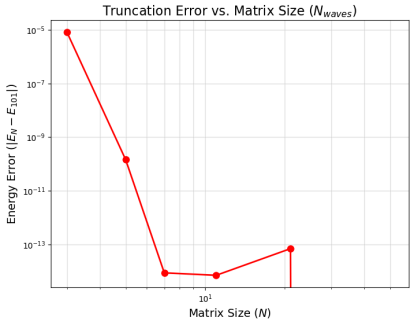

Figure 11 Matrix Solution

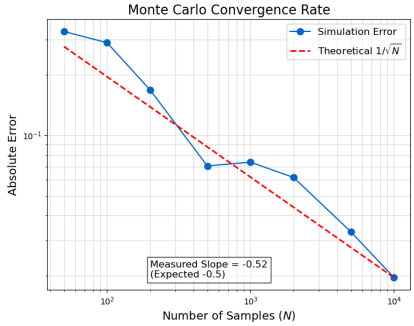

Figure 12 Monte Carlo Test

B. Discussion

改進：現在的問題是在我讓蒙地卡羅顯得很沒必要，我上次回到座位才想到可以就想說這代表我們可以隨機任取一個k，然後去得到一個能量，慢慢的就有一個能帶以及能隙結構的出現，也算是不錯的呈現;但言歸正傳我還是得加入一些測不准定理(energy vs t)來讓蒙地卡羅有點東西。但是我相信我有用蒙地卡羅視覺化能帶，畫出能隙以及“一群一群”的能帶，有畫出能帶結構，唯獨那些不確定性、機率都沒畫出效果，可惜唯獨時間不夠無法理解出怎麼視覺化符合 uncertainty principle。

References

[1] Kansas State University, *The Preparation of Dynamical Properties*, https://perg.phys.ksu.edu/vqmorig/tutorials/online/prep/.

[2] Kari Rummukainen, *Monte Carlo simulations in physics*, https://www.oulu.fi/tf/montecarlo/lectures/mc_notes1.pdf.